BIOSTATISTICS(BIO401)

MIDTERM SUBJECTIVE PREPARATION FALL 2023

BIOSTATISTICS

Statistics is a field of study that is concerned with:

• The collection, organization, summarization, and analysis of data.

• The drawing of inferences about a body of data, when only a part of data is observed.

Biostatistics is

• The use of statistic al tools for the data that is derived from biological sciences and medicine

Applications of Biostatistics:

- Public Health
- Pharmacology
- Epidemiology
- Medicine
- Genetics
- Genomics
- Proteomics
- Bioinformatics etc.

POPULATION

Population: In Statistics it is Defined as "an aggregate of entities for which we have an interest at a particular time." It can consists of Animals, machines, places or cells etc.

Example: If we are interested in knowing the weights of all the children enrolled in elementary schools of the Bahawalpur District. Then our population will be all the the children enrolled in elementary schools of the Bahawalpur District.

Types of Population

1. **Finite**

If a population of values consists of a fixed number of these values, the population is said to be finite. The finite population is also known as a countable population in which the population can be counted. In other words, it is defined as the population of all the individuals or objects that are finite. For statistical analysis, the finite population is more advantageous than the infinite population. ***Examples*** of finite populations are employees of a company, potential consumer in a market.

2. **Infinite**

If a population consists of an endless succession of values, the population is infinite one. The infinite population is also known as an uncountable population in which the counting of units in the population is not possible. ***Example*** of an infinite population is the number of germs in the patient's body is uncountable.

Some ***other concepts*** which are important to know:

- **Target Population:** The general population that the study seeks to understand.

- **Source population:** The specific individuals from which a representative sample will be drawn.
- **Sample Population:** Individuals ask to participate.
- **Study population:** Eligible participants.

Census:

A Census is a survey conducted on a full set observations belonging to a given population. It is defined as "*the complete enumeration of population of groups at a point in time with respect to well defined characteristics.*"

Population Parameter:

It is any summary number, like an average or percentage, that describes the entire population.

• These are the true values, which are usually unknown.

These values are denoted by Greek letters. e.g.: *Population Mean μ(Mu) ,Population Proportion π (Pi)*

SAMPLE

Sample is a representative part of a Population.

A **sample survey i**s a study that obtains data from a subset of a population of our interest, in order to estimate population attributes.

❖ **<u>Characteristics of a good sample</u>**

1) Representative.
2) It captures most of the variation in the population.
3) Obtained by using proper sampling methodology.
4) Focus on objective
5) Informative with minimum use of resources.

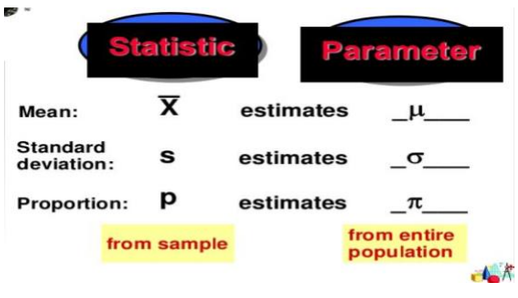
Difference b/w Population & Sample

POPULATION	SAMPLE
The measurable quality is called a parameter.	The measurable quality is called a statistic.
The population is a complete set.	The sample is a subset of the population.
Reports are a true representation of opinion.	Reports have a margin of error and confidence interval.
It contains all members of a specified group.	It is a subset that represents the entire population.

Sample Statistic

Any summary number, like an average or percentage, that describes the sample.

• Sample statistics are random variables. • These are the values which are usually denoted by Latin letters • *Sample Standard Deviation (s) • Sample Proportion (p).*



Types of Statistics

1. **Descriptive Statistics:**

Descriptive statistics are methods for organizing and summarizing data. This summarization can be in numeric as well as in graphical form. We use different methodologies within them. And these descriptive methodologies form the basis of any quantitative analysis of data.

Descriptive analysis is also known as Exploratory Data Analysis (EDA)

For example, tables or graphs are used to organize data, and descriptive values such as the average score are used to summarize data. *A descriptive value for a population is called a* ***parameter*** *and a descriptive value for a sample is called a* ***statistic****.*

2. **Inferential Statistics:**

These are the methods which we used to talk about the population from the information that comes out of the sample. ***OR***. These are statistical methodologies using which we reach a conclusion about a population on the basis of the information contained in the sample.

Inferential statistics are methods for using sample data to make general conclusions (inferences) about populations. Because a sample is typically only a part of the whole population, sample data provide only limited information about the population. As a result, sample statistics are generally imperfect representatives of the corresponding population parameters.

Inference

Use a random sample to learn something about a larger population

DATA

The raw material of statistics is called data. It is the collection of facts and figures. Which can occur as process of measurement or counting.

For example:

- When a hospital administrator counts the number of patients (counting).
- When a nurse weighs a patient (measurement)

Sources of Data

We search for suitable data to serve as the raw material for our investigation. Such data are available from one or more of the following sources:

1- Routinely kept records.

For example: - Hospital medical records contain immense amounts of information on patients.

-Hospital accounting records contain a wealth of data on the facility's business activities.

2- External sources.

The data needed to answer a question may already exist in the form of published reports, commercially available data banks, or the research literature, i.e. someone else has already asked the same question.

3- Surveys:

The source may be a survey, if the data needed is about answering certain questions.

For example: If the administrator of a clinic wishes to obtain information regarding the mode of transportation used by patients to visit the clinic, then a survey may be conducted among patients to obtain this information.

4- Experiments.

Frequently the data needed to answer a question are available only as the result of an experiment.

For example: If a nurse wishes to know which of several strategies is best for maximizing patient compliance, she might conduct an experiment in which the different strategies of motivating compliance are tried with different patients.

Types of Data

1. **Primary data r**efers to the first hand data gathered by the researcher himself. E.g: obtaining vitals of the patient directly by observing the patient(respondent)

. Primary data is the kind of data that is collected directly from the data source without going through any existing sources. It is mostly collected specially for a research project and may be shared publicly to be used for another research. Primary data is often reliable, authentic, and objective in as much as it was collected with the purpose of addressing a particular research problem. It is noteworthy that primary data is not commonly collected because of the high cost of implementation.

2. **Secondary data** means data collected by someone else earlier. e.g obtaining vitals of the patient from the available medical records.

Secondary data is the data that has been collected in the past by someone else but made available others to use. They are usually once primary data but become secondary when used by a third party. Secondary data are usually easily accessible to researchers and individuals because they are mostly shared publicly. This, however, means that the data are usually general and not tailored specifically to meet the researcher's needs as primary data does.

Variable

It is a characteristic that takes on different values in different persons, places, or things.

For example:

- heart rate,
- the heights of adult males,
- the weights of preschool children,
- the ages of patients seen in a dental clinic.

Types of Variable

- **Quantitative Variable:**

it is the one that can be measured and expressed numerically.

a) ***Discrete variable:*** it is the ones that have gap or jumps in the possible values. These gaps indicate the absence of values between particular possible values of our variable. It is also known as countable values. ***E.g.*** no of pages in book, teeth per child in a school, children in a family.
b) ***Continuous variable:*** It is the ones that take on each and every possible value within an interval. Don't possess gaps in their possible values. ***E.g.*** height of a person, weight of a person.

- **Qualitative variable:**

It is the one that can not be measure in a numerical form. These are non-numeric variable where each possible value is a category of a variable that's why they are also known as categorical variable or attribute. ***E.g.*** brand of PC, hair color, marital status, category of different types of animals etc.

Measurement

It is Defined as the assignment of numbers to objects or events according to a set of rule.

Properties of measurement scales:

Each scale of measurement satisfies one or more of the following properties of measurement.

☐ ***Identity***. Each value on the measurement scale has a unique meaning.

☐ ***Magnitude***. Values on the measurement scale have an ordered relationship to one another. That is, some values are larger and some are smaller.

☐ ***Equal intervals***. Scale units along the scale are equal to one another. This means, for example, that the difference between 1 and 2 would be equal to the difference between 19 and 20.

☐ ***A minimum value of zero.*** The scale has a true zero point, below which no values exist.

Types of Measurement Scales

There are four types of measurement scales • Nominal • Ordinal • Interval Data • Ratio

1) Nominal – Categorical data
2) Ordinal – shows ranks, intervals may vary
3) Interval – intervals are constant, arbitrary 0
4) Ratio – Numeric data with a 'real' 0 value.

Sampling Errors:

A sampling error is a deviation in the sample statistics value versus the true population parameter value. Sampling errors occur because the sample is not representative of the population or is biased in some way. Even randomized samples will have some degree of sampling error because a sample is only an approximation of the population from which it is drawn. If, for example , x is the mean obtained from a sample od size n and μ is the corresponding population parameter, then the difference between x and μ is sampling error, that is

$sampling\ error = x - \mu$

Categories

Descriptive statistics are also categorized into four different categories:

1) ***Measure of frequency:*** The frequency measurement displays the number of times a particular data occurs.
2) ***Measure of dispersion:*** Range, Variance, Standard Deviation are measures of dispersion.
3) ***Measure of central tendency:*** It identifies the spread of data. Central tendencies are the mean, median and mode of the data.
4) ***Measure of position:*** And the measure of position describes the percentile and quartile ranks.

Statistical inference can be divided into two main **branches**

Two ways to make inference

1) Estimation of parameters
    - Point Estimation [X(bar on x) or p]
    - Intervals Estimation
2) Hypothesis Testing
    - ***Null hypothesis:*** H0 Null Hypothesis states the Assumption to be tested e.g. SBP of participants = 120 (H0: u(meu) = 120).
    - ***Alternate hypothesis:*** H1 Alternative Hypothesis is the opposite of the null hypothesis (SBP of participants ≠ 120 (H1: u(mey ≠ 120). It may or may not be accepted and it is the hypothesis that is believed to be true by the researcher

Types of Sampling Methods

Ways to ensure representativeness: There are two possible ways to ensure that the selected sample is representative.

1) Random sample or probability sample:

The selection of units in the sample from a population is governed by the laws of chance or probability. The probability of selection of a unit can be equal as well as unequal.

2) **Non-random sample or purposive sample**:

The selection of units in the sample from the population is not governed by the probability laws. For example, the units are selected on the basis of the personal judgment of the surveyor. The persons volunteering to take some medical test or to drink a new type of coffee also constitute the sample on non-random laws.

❖ Probability Sampling Methods

This Sampling technique uses randomization to make sure that every element of the population gets an equal chance to be part of the selected sample. It's alternatively known as random sampling. Such sampling techniques can be classified in the following way.

• Simple random sampling • Stratified sampling • Systematic sampling • Cluster (area) sampling • Multistage sampling

TYPES

Simple Random Sampling: Every element has an equal chance of getting selected to be the part sample. It is used when we don't have any kind of prior information about the target population.

For example: Random selection of 20 students from class of 50 student. Each student has equal chance of getting selected. Here probability of selection is 1/50

Stratified Sampling: This technique divides the elements of the population into small subgroups (strata) based on the similarity in such a way that the elements within the group are homogeneous and heterogeneous among the other subgroups formed. And then the elements are randomly selected from each of these strata. We need to have prior information about the population to create subgroups.

Cluster Sampling: Our entire population is divided into clusters or sections and then the clusters are randomly selected. All the elements of the cluster are used for sampling. Clusters are identified using details such as age, sex, location etc. Cluster sampling can be done in following ways:

1) ***Single Stage Cluster Sampling:*** Entire cluster is selected randomly for sampling.

2) ***Two Stage Cluster Sampling:*** Here first we randomly select clusters and then from those selected clusters we randomly select elements for sampling

Systematic Clustering: Here the selection of elements is systematic and not random except the first element. Elements of a sample are chosen at regular intervals of population. All the elements are put together in a sequence first where each element has the equal chance of being selected.

Multi-Stage Sampling It is the combination of one or more methods described above. Population is divided into multiple clusters and then these clusters are further divided and grouped into various sub groups (strata) based on similarity. One or more clusters can be randomly selected from each stratum. This process continues until the cluster can't be divided anymore. For *example* country can be divided into states, cities, urban and rural and all the areas with similar characteristics can be merged together to form a strata.

❖ Non-Probability Sampling Methods

It does not rely on randomization. This technique is more reliant on the researcher's ability to select elements for a sample. Outcome of sampling might be biased and makes difficult for all the elements of population to be part of the sample equally. This type of sampling is also known as non-random sampling.

• Convenience Sampling • Purposive Sampling • Quota Sampling • Referral /Snowball Sampling

TYPES

Convenience Sampling: Here the samples are selected based on the availability. This method is used when the availability of sample is rare and also costly. So based on the convenience samples are selected. For *example*: Researchers prefer this during the initial stages of survey research, as it's quick and easy to deliver results.

Purposive Sampling This is based on the intention or the purpose of study. Only those elements will be selected from the population which suits the best for the purpose of our study. For Example: If we want to understand the thought process of the people who are interested in pursuing master's degree then the selection criteria would be "Are you interested for Masters in..?" All the people who respond with a "No" will be excluded from our sample.

Quota Sampling This type of sampling depends of some pre-set standard. It selects the representative sample from the population. Proportion of characteristics/ trait in sample should be same as population. Elements are selected until exact proportions of certain types of data is obtained or sufficient data in different categories is collected

For example: If our population has 45% females and 55% males then our sample should reflect the same percentage of males and females.

Referral /Snowball Sampling This technique is used in the situations where the population is completely unknown and rare. Therefore we will take the help from the first element which we select for the population and ask him to recommend other elements who will fit the description of the sample needed. So this referral technique goes on, increasing the size of population like a snowball.

For *example*: It's used in situations of highly sensitive topics like HIV Aids where people will not openly discuss and participate in surveys to share information about HIV Aids. Not all the victims will respond to the questions asked so researchers can contact people they know or volunteers to get in touch with the victims and collect information.

Describing Data With Tables

Descriptive Statistics is performed using ***Tabular, Graphical, and Numerical methods.***

- **Tabular Methods** are used to summarize the data in table form. It is a systematic organization of information in grid row and columnar structure. The most frequently used tabular format for data summarization is Frequency table and Cross-tabulation.
- **Graphical Methods** are a visual way of presenting data using charts and graphs. The visuals make the data intuitive and self-understandable. The most frequently used visual representation of data are Bar Plot, Histogram, Pareto Chart, Box Plot, Pie Chart, Line Plot, and Scatter Plot.

Ordered array

It is the listing of the values of a collection(either sample or population) in order of magnitude from the smallest value to largest value.

Statistical Distribution

It is the distribution of statics across an infinite number of samples

Frequency Distribution

It is the organization of Raw data into tabular form using two important components of frequency distribution.

- Class. And
- Class frequency

Class is a quantitative or qualitative variable whereas ***class frequency*** is the number of data values in a specific class.

Class is a range for the values of a variable. ***Frequency*** is the number of observations associated with a class. ***Relative Frequency*** is the proportion of observations (frequency) associated with a class

Frequency distribution

. A table that organizes data values into classes or intervals along with number of values that fall in each class (frequency, f).

<u>Types of frequency distribution</u>

- ***Categorical frequency Distribution*** : is used for qualitative data {Nominal & Ordinal}
- ***Ungrouped frequency distribution:*** is used for quantitative data { Interval & Ratio} for data sets with few different values, each value is in its own class
- ***Grouped frequency distribution:*** is used for quantitative data. It shows the frequency of an item in each separate data value rather than groups of data values. for data sets with many different values, which are grouped together in the classes.

Relative frequency

• Shows the portion or percentage of the data that falls in a particular class. OR

A relative frequency is the ratio (fraction or proportion) of the number of times a value of the data occurs in the set of all outcomes to the total number of outcomes.

To convert a frequency into a proportion or relative frequency, we should divide the frequency for each class by the total of the frequencies. The sum of the relative frequencies will always be 1.

R.F= frequency of the class/total

Cumulative relative frequency

It is the accumulation of the previous relative frequencies. To find the cumulative relative frequencies, add all the previous relative frequencies to the relative frequency for the current row.

Cumulative relative frequency = sum of previous relative frequencies + current class frequency

Range

The range which is defined as the difference between the largest value and the smallest value.

Range = Xm(maximum value) – XO(minimum value)

Class Interval

Divide the range by the chosen number of classes in order to obtain the approximate value of the class interval i.e. the width of our classes. Class interval is usually denoted by ***h***.

h= range/classes

Contingency tables

Contingency tables (also called crosstabs or two-way tables) are used in statistics to summarize the relationship between several categorical variables. A contingency table is a special type of frequency distribution table, where two variables are shown simultaneously. The entries in the cells of a two-way table can be frequency counts or relative frequencies (just like a one-way table).

Describing Data With Charts

What is data?

The concept of data as it is used in the syllabus is commonly referred to as 'raw' data – a collection of text, numbers and symbols with no meaning. Data therefore has to be processed, or provided with a context, before it can have meaning.

What is information?

It is important that students learn the concept of what 'information' is as used in information technology. Information is the result of processing data, usually by computer. This results in facts, which enables the processed data to be used in context and have meaning. Information is data that has meaning.

What is knowledge?

When someone memorizes information, this is often referred to as 'rote-learning' or 'learning by heart'. We can then say that they have acquired some knowledge. Another form of knowledge is produced as a result of understanding information that has been given to us, and using that information to gain knowledge of how to solve problems.

How are data, information and knowledge linked?

If we put Knowledge into an equation it would look like this: Information + application or use = Knowledge

Types of Data: Cross Sectional and Time Ordered Cross-sectional Data

- ***Cross-section data*** is collected in a single time period and is characterized by individual units - people, companies, countries, etc.
- ***Time Series Data*** collected at a number of specific points in time is called time series data. Such examples include stock prices, interest rates, exchange rates as

well as product prices, GDP, etc. Time series data can be observed at many different frequencies (hourly, daily, weekly, monthly, quarterly, annually, etc.). Unlike cross-sectional data, the ordering of the data is important in time-series data.

Frequency Histograms

A graphical display of distribution of frequencies is known as frequency histogram.

A bar graph that represents the frequency distribution.

- The horizontal scale is quantitative and measures the data values.
- The vertical scale measures the frequencies of the classes.
- Consecutive bars must touch.

Bar and Pie Charts

Frequency tables, pie charts, and bar charts can be used to display the distribution of a single categorical variable. These displays show all possible values of the variable along with either the frequency (count) or relative frequency (percentage).

Pie Chart: A circle is divided into sectors That represent categories.

Pareto Chart(Bar): • A vertical bar graph in which the height of each bar represents frequency or relative frequency.

Scatter Plot

Scatter plots (also called scatter graphs) are similar to line graphs. A line graph uses a line on an X-Y axis to plot a continuous function, while a scatter plot uses dots to represent individual pieces of data. In statistics, these plots are useful to see if two variables are related to each other. For example, a scatter chart can suggest a linear relationship (i.e. a straight line). Scatter plots are also called scatter graphs, scatter charts, scatter diagrams and scatter grams.

Time series plot

A time series graph is a line graph of repeated measurements taken over regular time intervals. Time is always shown on the horizontal axis. On time series graphs data points are drawn at regular intervals and the points joined, usually with straight lines. Time series graphs help to show trends or patterns.

Box and whisker plot

It shows five number summary of the set of data: *minimum, lower quartile, median, upper quartile, maximum.*

A box plot (also called a box and whisker plot) shows data using the middle value of the data and the quartiles, or 25% divisions of the data.

Construction:

Step 1: Arrange the data in ascending order.

Step 2: Find the median, lower quartile and upper quartile.

Step 3: Draw a number line that will include the smallest and the largest data.

Step 4: Draw three vertical lines at the lower quartile (12), median (22) and the upper quartile (36), just above the number line.

Step 5: Join the lines for the lower quartile and the upper quartile to form a box.

Step 6: Draw a line from the smallest value (5) to the left side of the box and draw a line from the right side of the box to the biggest value (53).

Describing Data From Shapes

What are outlier?

Unusual data values as compared to the rest of the set. They may be distinguished by gaps in a histogram.

What are shapes of distribution?

Symmetric • Data is symmetric if the left half of its histogram is roughly a mirror image of its right half.

Skewed • Data is skewed if it is not symmetric and if it extends more to one side than the other.

Uniform • Data is uniform if it is equally distributed (on a histogram, all the bars are the same height or approximately the same height).

HISTOGRAM

A histogram consists of a set of adjacent rectangles whose bases are marked off by class boundaries along the X-axis, and whose heights are proportional to the frequencies associated with the respective classes.

FREQUENCY POLYGON:

A frequency polygon is obtained by plotting the class frequencies against the mid-points of the classes, and connecting the points so obtained by straight line segments.

FREQUENCY CURVE: When the frequency polygon is smoothed, we obtain what may be called the frequency curve.

Besides Histograms, there are other methods of graphing quantitative data:

- Stem and Leaf Plots
- Dot Plots
- Time Series

Stem and Leaf Plots

Represents data by separating each data value into two parts: the stem (such as the leftmost digit) and the leaf (such as the rightmost digit).

Dot plot

• Consists of a graph in which each data value is plotted as a point along a scale of values.

Central Tendency

A measure of central tendency is a measure which indicates where the middle of the data is. The three most commonly used measures of central tendency are:

1) The ***Mean***: It is the average of the data.
2) The ***Median***: When ordering the data, it is the observation that divide the set of observations into two equal parts such that half of the data are before it and the other are after it.
3) The ***Mode***: It is the value which occurs most frequently. If all values are different there is no mode. Sometimes, there are more than one mode.

VARIOUS TYPES OF AVERAGES:

- The arithmetic mean,
- The geometric mean,
- The harmonic mean
- The median, and

- The mode

Quantiles

Location parameter

The mean and median are special cases of a family of parameters known as location parameter.

Measures of position

Other location parameters which helps us to determine the position of a particular observations in a data are called measures of position.

Quantile

These measures of position when help us to divide sample data into **n** equal part or divide the probability distribution into contiguous intervals with equal probabilities are called quantiles.

Types of quantiles

1) **Quartiles:** In an array quartiles are 3 cut points which divide the values into 4 equal parts. Denoted by Qk(k is a subscript).
2) **Deciles:** In an array the Deciles are 9 values/cut-points that divides the values into 10 equal parts. Denoted by Dk
3) **Percentile:** In an array the Percentiles are 99 values/cut-points that divides the values into 100 equal parts. Denoted by Pk

Interquartile range

The interquartile range is the difference between the upper quartile and the lower quartile.

IQR= Upper quartile – Lower Quartile

FORMULA

THE ARITHMETIC MEAN: "The arithmetic mean or simply the mean is a value obtained by dividing the sum of all the observations by their number."

For Ungrouped Data:

$$\overline{X} = \frac{\sum_i X_i}{n}$$

For Grouped Data: In case of Frequency Distribution

$$\overline{X} = \frac{\sum fx}{\sum f}$$

MEDIAN

For Grouped Data:

$$Median = L + \left(\frac{N}{2} - C.F.\right)\frac{h}{f}$$

For Ungrouped Data:

For odd: n+1/2

For even: (n+1/2) + (n+1/2) /2

Types of Measures of Dispersion

There are two main types of dispersion methods in statistics which are:

• Absolute Measure of Dispersion

- Range= Xmax – Xmin(R = Xm – X0 where Xm represents the highest value and X0 the lowest.)
- Variance (σ^2) = $\sum(X-\mu)^2/N$
- Standard Deviation (S.D) = $\sqrt{\sigma}$
- Quartiles and Quartile Deviation: Q3 - Q1
- Mean and Mean Deviation:

• Relative Measure of Dispersion

1) Co-efficient of Range
2) Co-efficient of Variation
3) Co-efficient of Standard Deviation
4) Co-efficient of Quartile Deviation
5) Co-efficient of Mean Deviation

The common coefficients of dispersion are:

C.D. in terms of	Coefficient of dispersion
Range	C.D. = $(X_{max} - X_{min})/(X_{max} + X_{min})$
Quartile Deviation	C.D. = $(Q3 - Q1)/(Q3 + Q1)$
Standard Deviation (S.D.)	C.D. = S.D./Mean
Mean Deviation	C.D. = Mean deviation/Average

MEAN DEVIATION

MEAN DEVIATION

$$M.D = \frac{\sum |X - \bar{X}|}{n} \quad \text{(for ungrouped data)}$$

$$M.D = \frac{\sum f\,|X - \bar{X}|}{\sum f} \quad \text{(for grouped data)}$$

VARIANCE

For ungrouped data.

Variance can be calculated as for ungrouped data.

$$Var = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$$

THE Variance IN CASE OF GROUPED DATA

Formula

$$Var = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2$$

STANDARD DEVIATION

$$S = \frac{\sum \left(X - \bar{X}\right)^2}{n}$$

For ungrouped data.

Standard deviation can be calculated as for ungrouped data.

$$S.D = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

THE STANDARD DEVIATION IN CASE OF GROUPED DATA

Formula

$$S.D = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

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